

Solving Hamiltonian Constraint Equation with Physics Informed Neural Operators

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Y.-C. Zhou *et al.*, “Solving Hamiltonian Constraint Equation with Physics-Informed Neural Networks,” *Phys. Rev. D*, Jun. 2026, doi: [10.1103/619s-p6md](https://doi.org/10.1103/619s-p6md).

Z. Li *et al.*, “Physics-Informed Neural Operator for Learning Partial Differential Equations,” Jul. 29, 2023, *arXiv*: arXiv:2111.03794. doi: [10.48550/arXiv.2111.03794](https://doi.org/10.48550/arXiv.2111.03794).

G. B. Cook, “Initial Data for Numerical Relativity,” *Living Rev. Relativ.*, vol. 3, no. 1, p. 5, Dec. 2000, doi: [10.12942/lrr-2000-5](https://doi.org/10.12942/lrr-2000-5).

Background

- Processing of gravitational wave data detected on Earth requires accurate snapshot of initial gravitational field data
- Focusing on non-spinning binary blackhole (BBH) systems
- Traditional numerical relativity (NR) techniques to examine compact binary object coalescence (CBC) are computationally expensive
 - ML methods can be more computationally efficient

Theory (1)

Gravitational wave evolution governed by Einstein's field equations

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

- 6 evolution equations
- 4 constraint equations

Constraint equations can be reformulated

- Hamiltonian constraint $\bar{R} + K^2 - K_{ij}K^{ij} = 16\pi G\rho$
- Momentum constraint $\bar{\nabla}_j(K^{ij} - \gamma^{ij}K) = 8\pi G j^i$

Theory (2)

Lichnerowicz-York Conformal Method

$$\Delta\psi + \frac{1}{8}\psi^{-7}\bar{K}_{ij}\bar{K}^{ij} = 0$$

Decomposing conformal factor ψ

$$\psi = 1 + \sum_n \frac{m_n}{2r_n} + u$$

$$\Delta u + \frac{1}{8} \left(1 + \sum_n \frac{m_n}{2r_n} + u \right)^{-7} \bar{K}_{ij} \bar{K}^{ij} = 0$$

Zhou et. al (PINN)

MLP architecture:

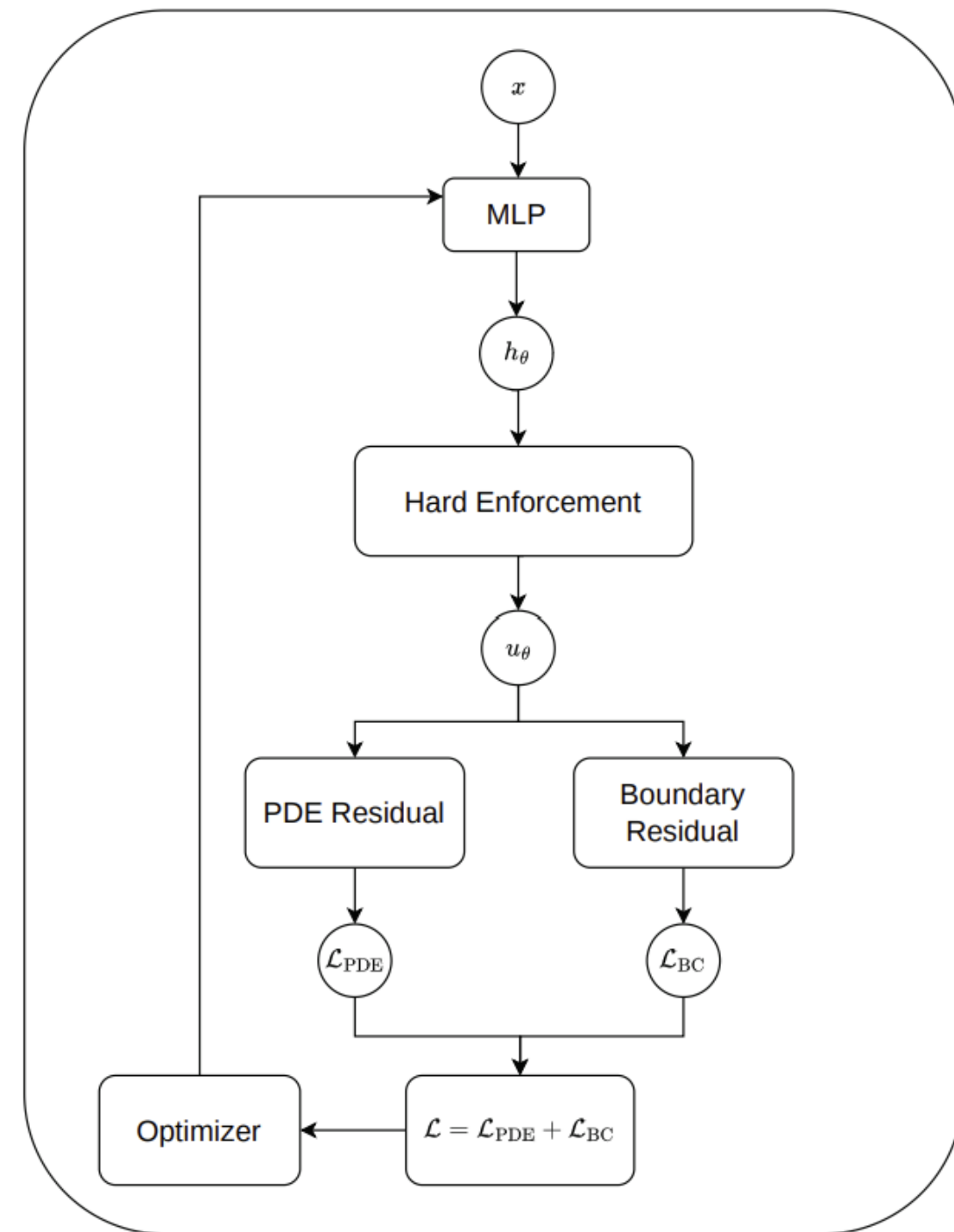
- 3 hidden layers (64 neurons each)
- SiLU activation function
- Adam optimizer $\text{lr} = 3 \times 10^{-4}$
- 5000 epochs

Guided hard-enforcement ansatz:

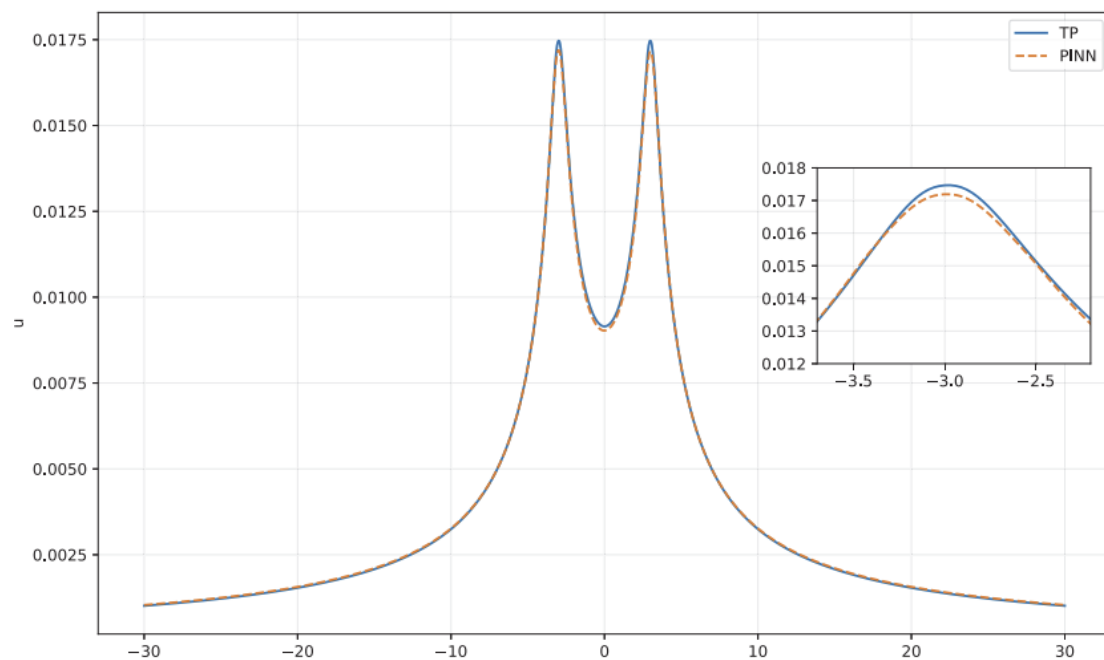
$$u_{\theta}(x) = \kappa u_g(x) [1 + cW(x) \tanh(h_{\theta}(x))]$$

$$\mathcal{L}_{PDE} = w_2 L_2 + w_{\infty} \text{soft} - L_{\infty}$$

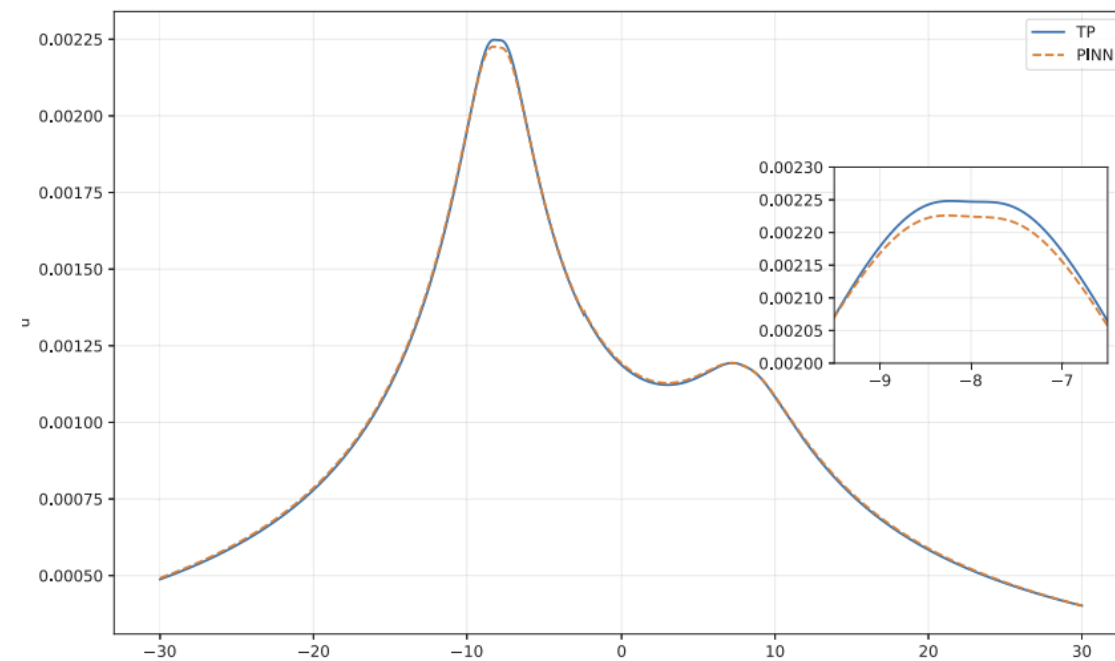
$$\mathcal{L}_{BC} = w_{rob} \frac{1}{N_{d\Omega}} \sum_{i=1}^{N_{d\Omega}} \mathcal{R}_{B,i}^2$$



Results



$$\text{L2RE} = 0.0172 \pm 0.0040$$

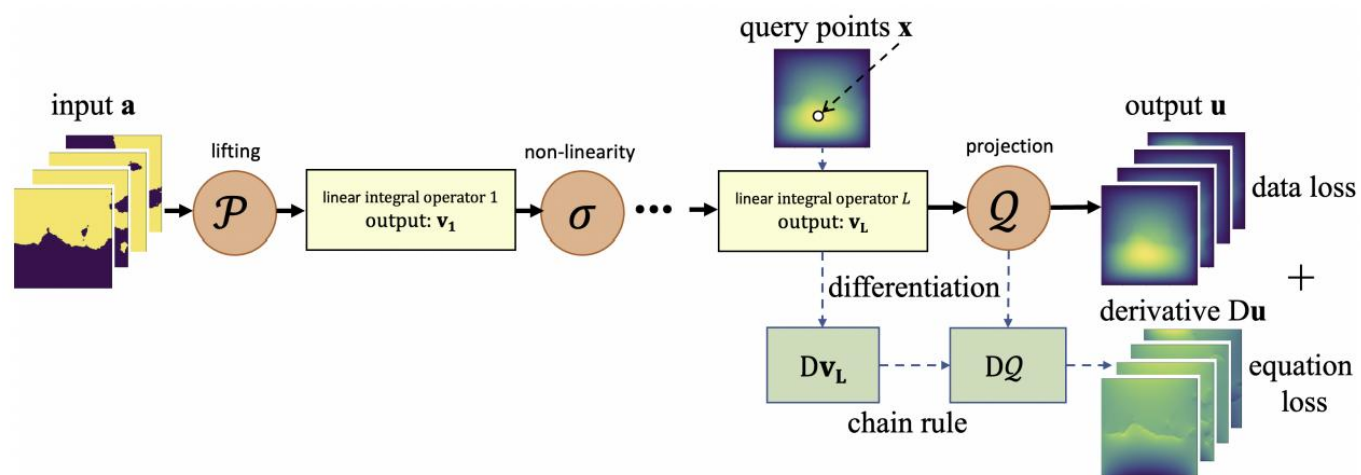


$$\text{L2RE} = 0.0077 \pm 0.0007$$

Project Proposal (1)

Train a Physics Informed Neural Operator (PINO) to generate initial data for non-spinning BBH systems

$$\text{PINO} = \text{FNO} + \text{PINN}$$



Project Proposal (2)

- Parameterise relative masses of the blackholes
- Use same ansatz and loss functions developed by Zhou et. al
- Train on data generated from TwoPunctures program
- Predict gravitational field correction for unseen relative masses

Setup:

- Based on simple non-spinning equal-mass BBH system in Zhou et. al paper
- Blackholes 6 units of space apart orbiting centre of mass (CoM)
- Momentum of the blackholes scaled by relative masses

Project Plan

1. Recreate the 2 non-spinning BBH system results from the Zhou et. al paper
2. Train PINO for $m_1 = 0.1, 0.2, 0.3, 0.4, 0.5$
3. Predict u_θ for $m_1 = 0.25, 0.35$
4. Validate against initial data generated from TwoPunctures program

Extension:

- Test zero-shot super resolution capability of PINO
- Train on coarse data from TwoPunctures
- Predict u_θ for higher-resolution grid

Thank you for listening

Suggestions?